Avon Means River

The old roots and everyday words of the river names of the United Kingdom, from OpenStreetMap

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Abstract

River names are among the oldest words in the British landscape. I join 128,210 named OpenStreetMap waterway stretches in the United Kingdom into 32,068 named watercourses, split each name into a generic word (river, afon, allt, burn, beck, brook and others) and a specific name, and match the specific names against sixteen groups of Brittonic or older roots after Ekwall's *English River-Names*, assigning every stretch of the 116 rivers in those groups to its river by hand. Of the 60 largest English rivers in the data, 37 have names generally traced to Brittonic or older sources, 10 have English names (several of them late back-formations from towns) and 13 are debated, while 72.7% of English towns have a first word ending in a common English or Norse element. There are 10 Avons, 10 Esks and 15 Calders. The generic words map the old language frontiers: 86.3% of Welsh streams named with *nant*, *afon*, *brook* or *river* use one of the two Welsh words; Gaelic *allt* and *abhainn* make up 86.2% of the four main Scottish generics in Highland and none in the Borders; and Old Norse *beck* forms the majority north of about 53.75°N in England. At similar latitudes, 81.6% of beck and brook names in Lincolnshire and Nottinghamshire are becks, against 0.0% in Cheshire, Staffordshire, Shropshire and Greater Manchester; Derbyshire and Leicestershire, also within the Danelaw, have almost none.

Keywords: river names; hydronymy; toponymy; Brittonic; Old Norse; Danelaw; Gaelic; Welsh; OpenStreetMap

1 Introduction

It is a commonplace of English place-name study that river names are older than settlement names: the Anglo-Saxons renamed most farms and villages in English but kept many British names for rivers, so that *Avon*, the Brittonic word for a river (Welsh *afon*), became the name of several (Ekwall, 1928; Gelling and Cole, 2000; Mills, 2011). Ekwall's *English River-Names* remains the fullest survey of the names themselves (Ekwall, 1928). This paper does not re-examine any etymology. It asks what the standard readings look like when applied to every named river and stream in a modern open map: how many rivers share the old roots, how England's largest rivers compare with its towns, and how the everyday generic words (burn, beck, brook, allt, nant) divide the country.

2 Data

Waterways. OpenStreetMap ways tagged `waterway=river`, `stream` or `canal` in the United Kingdom with a name, queried with QLever (Bast and Buchhold, 2017): 128,210 named ways with `name`, `name:cy` and `name:gd` and the first coordinate of each way. **Places.** OpenStreetMap place nodes (city, town, village, hamlet) in the UK, with county and nation from the ONS boundaries.

3 Method

From ways to watercourses. Ways with the same normalised name and type were joined when their start points chain within 15 km (rivers), 8 km (canals) or 4 km (streams), giving 32,068 watercourses (4,048 rivers, 27,153 streams, 867 canals; canals are left out of everything below). Because OpenStreetMap splits one river into pieces with gaps, different tags or a Welsh name, and because two rivers of one name may lie close, every piece of the rivers in the root groups was assigned to its river by hand from its position (`code/root_rivers_by_hand.csv`).

Generic and specific. Each name was split into a generic (before the name: River, Afon, Allt or Alltan, Abhainn, Nant, Water of, Burn of; after it: Brook, Beck, Burn, Bourne, Water, Gill or Ghyll, Sike or Syke, Dyke or Dike, Rhyne, Lode, Fleet, River; or a one-word name ending in one of these) and a specific name, with leading qualifiers (Great, Little, North, Black and so on) removed for matching.

Roots. Sixteen root groups (Table 2) after Ekwall (1928), with Watts (2004) and Mills (2011), are matched to the specific name or the Welsh name tag. Matches were read by eye; 22 were removed by hand (for example Kent Water and Kent Ditch, named after the county, and several minor *Allen* and *Ouse* streams named after people or places).

Largest rivers and towns. The 60 largest English rivers are ranked by the spread of their mapped start points (a proxy for length) and each name is classed as Brittonic or older, English (including back-formations), or uncertain after Ekwall (1928) and Watts (2004) (Table 1). English towns and cities are classed by whether their first word ends in a common English or Norse element (*-ton*, *-ham*, *-ley*, *-bury*, *-ford*, *-by*, *-thorpe*, *-wick*, *-field* and others); this is a crude test that counts some older names ending in these letters and misses English names with other endings.

4 Results

4.1 Rivers and towns

37 of the 60 largest English rivers (61.7%) have names traced to Brittonic or older sources, 10 (16.7%) English names and 13 uncertain ones; 9 of the largest ten are pre-English (Figure 1). Of 1,184 English towns and cities, 72.7% have a first word with an English or Norse ending, and of 11,671 villages 71.5%. Within 6 km of 45 rivers with old names, 123 towns and villages carry the river's name, such as Exeter, Dartmouth, Alnwick, Tynemouth and Doncaster; a few rivers were instead named after their towns, so the direction must be checked case by case.

Table 1: The 60 largest English rivers in the data, by the spread of their mapped course (km, a proxy for size), and the origin of the name after Ekwall (1928) and Watts (2004).

Rank	River	Spread	Origin	Note
1	Thames	184	Brittonic or older	
2	Trent	107	Brittonic or older	
3	Nene	106	Brittonic or older	
4	Ouse	100	Brittonic or older	
5	Avon	95	Brittonic or older	
6	Welland	86	uncertain	
7	Severn	77	Brittonic or older	
8	Derwent	75	Brittonic or older	
9	Ure	75	Brittonic or older	

Table 1: The 60 largest English rivers in the data, by the spread of their mapped course (km, a proxy for size), and the origin of the name after Ekwall (1928) and Watts (2004).

Rank	River	Spread	Origin	Note
10	Teme	73	Brittonic or older	
11	Wye	72	Brittonic or older	
12	Wharfe	68	uncertain	
13	Calder	67	Brittonic or older	
14	Ribble	65	uncertain	
15	Kennet	64	Brittonic or older	
16	Tamar	58	Brittonic or older	
17	Exe	58	Brittonic or older	
18	Stour	56	uncertain	
19	Mersey	55	English	Old English, the boundary river
20	Taw	53	Brittonic or older	
21	Aire	52	Brittonic or older	
22	Yare	52	Brittonic or older	
23	Lune	51	Brittonic or older	
24	Waveney	51	English	Old English, river by a quaking fen
25	Don	50	Brittonic or older	
26	Witham	49	uncertain	
27	Medway	49	Brittonic or older	
28	Wey	49	Brittonic or older	
29	Lea	48	Brittonic or older	
30	Tees	47	Brittonic or older	
31	Parrett	46	uncertain	
32	Cherwell	46	English	Old English wella, stream; first element uncertain
33	Lugg	45	Brittonic or older	
34	Dove	45	Brittonic or older	
35	Brue	41	Brittonic or older	
36	Test	41	uncertain	
37	Soar	41	Brittonic or older	
38	Colne	39	Brittonic or older	
39	Weaver	39	English	Old English, winding river
40	Lark	38	English	back-formation from Lackford
41	Thame	38	Brittonic or older	
42	Rother	37	English	back-formation from Rotherfield (the eastern Rother; the western Rother is from Rotherbridge)
43	Swale	37	uncertain	
44	Frome	37	Brittonic or older	
45	Bure	37	uncertain	
46	Tweed	37	Brittonic or older	
47	Wissey	36	uncertain	
48	Tern	35	uncertain	
49	Tame	35	Brittonic or older	
50	Torridge	35	uncertain	
51	Dane	34	uncertain	
52	Arun	34	English	back-formation from Arundel
53	Tyne	34	Brittonic or older	
54	Roding	33	English	from the Roding villages, Old English folk name
55	Wear	33	Brittonic or older	
56	Windrush	33	Brittonic or older	
57	Mole	33	English	back-formation from Molesey

Table 2: Root groups, the meaning given in the literature, the reading, and the number of distinct rivers and streams in the UK data.

Names	Root and meaning	Reading	Rivers
Avon	Brittonic *abona, 'river' (Welsh afon)	certain	10
Esk, Exe, Axe, Usk	Brittonic *isca, usually read as 'water'	probable	16
Ouse	a Brittonic or older word for 'water'	probable	5
Derwent, Darwen, Darent, Dart	Brittonic *derwā, 'oak': 'river where oaks grow'	certain	11
Dee	Brittonic *dēwā, 'goddess, holy one'	probable	4
Don	perhaps the goddess Dānu, or a word for water	probable	5
Thames, Tame, Thame, Teme, Tamar, Team	a pre-English root *tam-, often read as 'dark'	probable	8
Calder	Brittonic *caleto-dubron, 'hard (rocky) water'	certain	15
Leven	Brittonic *lēβn-, 'smooth'	probable	5
Aln, Alne, Allen, Ellen, Alun	a pre-English root *alaun-, meaning uncertain	probable	15
Frome	Brittonic *frām, 'fair, fine, brisk'	probable	6
Kennet, Kent	a pre-English root of uncertain meaning (Roman Cunetio)	probable	2
Colne, Clun	pre-English, meaning unknown	probable	5
Wye, Wey	pre-English, meaning unknown	probable	5
Trent	Brittonic, often read as 'the trespasser', the river that floods	probable	1
Severn	pre-English (Latin Sabrina, Welsh Hafren), meaning unknown	certain	3

Table 1: The 60 largest English rivers in the data, by the spread of their mapped course (km, a proxy for size), and the origin of the name after Ekwall (1928) and Watts (2004).

Rank	River	Spread	Origin	Note
58	Tone	33	Brittonic or older	
59	Otter	32	English	Old English otor, otter
60	Kent	32	Brittonic or older	

4.2 The roots

Table 2 lists the root groups and the number of distinct rivers and streams in each. There are 10 Avons (6 in England, 4 in Scotland, where some may be Gaelic rather than Brittonic), 10 Esks, 6 Derwents, Darwens and Darents, and 15 Calders. In all 116 rivers and streams fall in the sixteen groups (Figures 2 and 3).

4.3 The generic words

Burn (5,792), allt (4,263), brook (3,284), beck (2,969), gill (1,760), nant (1,480), sike (1,136), water (825) and afon (793) are the commonest generics (Figure 4). In Wales (Owen and Morgan, 2007) 86.3% of streams named with *nant*, *afon*, *brook* or *river* are *nant* or *afon*. In Scotland (Watson, 1926) Gaelic *allt* and *abhainn* account for 47.9% of the four main generics overall, 86.2% in Highland, 98.8% in the Western Isles, 16.7% in Aberdeenshire and none in the Scottish Borders or Dumfries and Galloway, where the streams are burns, waters and sikes (the Gaelic past of Galloway shows



Figure 1: The 60 largest English rivers in the data and the origin of their names.

instead in names in *Pol-*, from Gaelic *poll*, such as Polmaddy Burn). Local words follow local landscapes: 26 rhynes in Somerset and 8 lodes in Cambridgeshire.

4.4 The beck line

Becks outnumber brooks in England north of about 53.75°N (Figure 5). The 5th percentile of beck latitudes is 53.56°N and the 95th percentile of brook latitudes 53.74°N, so the change is sharp. It is not only latitude: in Lincolnshire and Nottinghamshire (median place latitude 53.19°N) there are 115 becks to 26 brooks (81.6% becks), against 0 becks to 755 brooks in Cheshire, Staffordshire, Shropshire and Greater Manchester (52.90°N). Lincolnshire (96.0%) and Norfolk (94.7%) are almost all beck. Derbyshire and Leicestershire, within the Danelaw of the Five Boroughs, have 2 becks to 300 brooks, so the beck tracks dense Norse-speaking settlement rather than the political boundary.

5 Limitations

The etymologies are the standard ones, applied without re-examining early spellings; several are debated (the meaning of the *Tam-* root and of **isca*, the Scottish Avons, Stour, Wharfe, the first element of Cherwell) and the origin classes of the 60 largest rivers should be read as a summary of



Figure 2: The courses of the rivers in the sixteen root groups.

the literature, not a new judgement. OpenStreetMap's naming of small streams is uneven between regions and mappers, and many small streams are unnamed in it; the generic shares describe named streams in the map, not the whole historical record. The size ranking is a proxy: rivers mapped in a few long ways score low, so some long rivers, such as the Eden in Cumbria, fall outside the 60, and the River Tweed counts only for its English stretch. A ranking by true length would change the list at the margin. The town test reads only word endings. The joining of ways into rivers was checked by hand only for the root groups.

6 Conclusion

Mapped from an open gazetteer, Britain's river names show both layers that the place-name literature describes: a scatter of very old names, Avon, Esk, Derwent, Calder, Dee, under an English landscape of towns, and an everyday vocabulary of streams that still follows the old language



Figure 3: Avon; Esk, Exe, Axe and Usk; Derwent and Dart; and Calder.

boundaries, Gaelic in the Highlands, Welsh in Wales, and the Norse beck where Scandinavian settlers were thickest.

Data and code availability

The interactive map is at https://maps.tristanhotham.com/river-names/. Code, the root rules, the hand assignments and the derived tables are at https://github.com/Tristan-Hotham/river-names and archived with this paper at https://doi.org/10.5281/zenodo.24063209. The river tables are derived from OpenStreetMap (© OpenStreetMap contributors) and released under the Open Database Licence; text and figures CC BY 4.0; code MIT. County boundaries from the ONS (Open Government Licence v3.0).

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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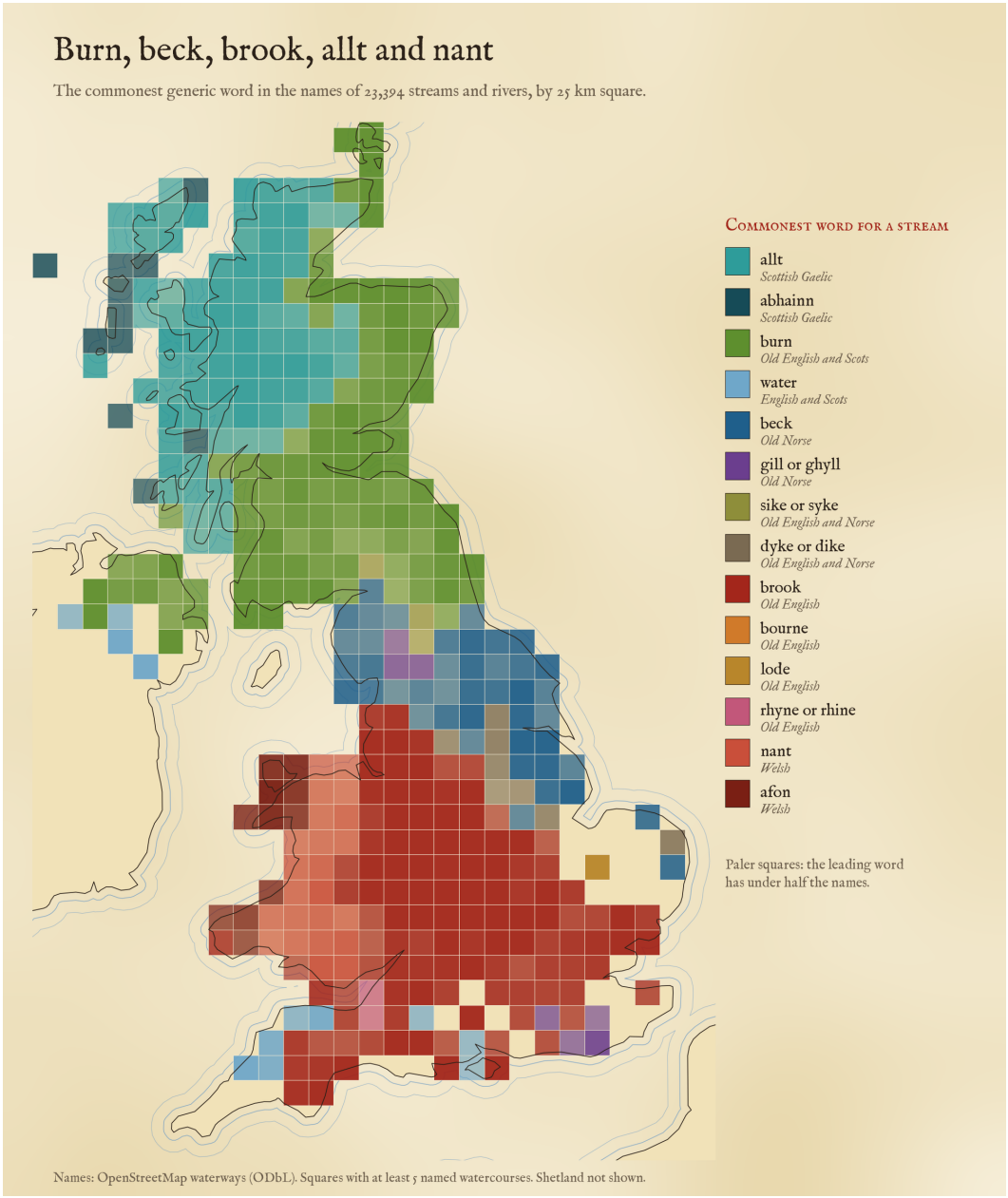


Figure 4: The commonest generic word in each 25 km square (at least five named watercourses).

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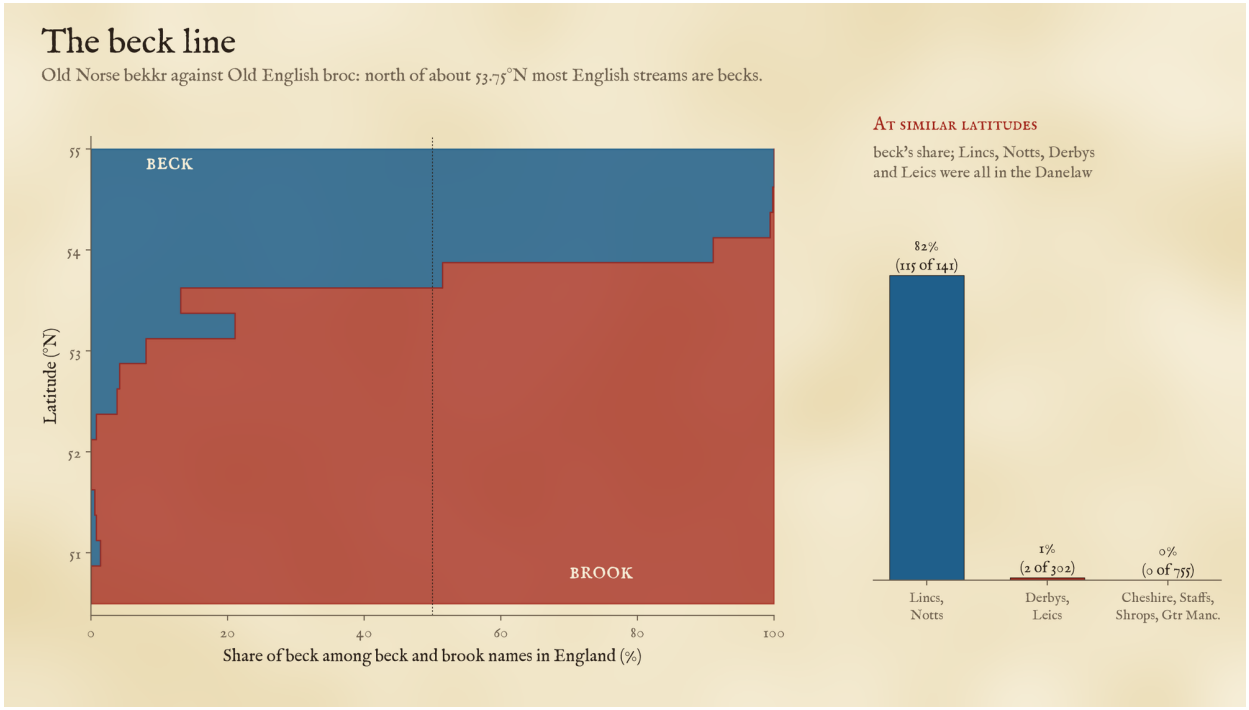


Figure 5: Beck against brook by latitude in England (left) and in eastern and western counties at similar latitudes (right).